

LAB EXPERIMENT RECORD

SET 13 — Geographic Data Analysis

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursday

A. Aim

To analyze geographic data by creating a map chart, scatter plot, and data table using R, and to develop an interactive Tableau dashboard combining these visualizations.

B. Dataset

Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Program 1 — Map Chart

Aim

To create a map chart to visualize the distribution of cities on a geographic map and label the map elements.

R Code

```
# Geographic Data
```

```
city <- c(  
  "City A",  
  "City B",  
  "City C"  
)
```

```
population <- c(  
  500000,  
  700000,  
  600000  
)
```

```
temperature <- c(  
  75,  
  68,  
  80  
)
```

```
elevation <- c(  
  1000,  
  800,
```

```
1200
)

# Sample coordinates for map demonstration

latitude <- c(
  13.08,
  13.15,
  13.00
)

longitude <- c(
  80.27,
  80.30,
  80.25
)

geo_data <- data.frame(
  City = city,
  Population = population,
  Avg_Temperature = temperature,
  Elevation = elevation,
  Latitude = latitude,
  Longitude = longitude
)

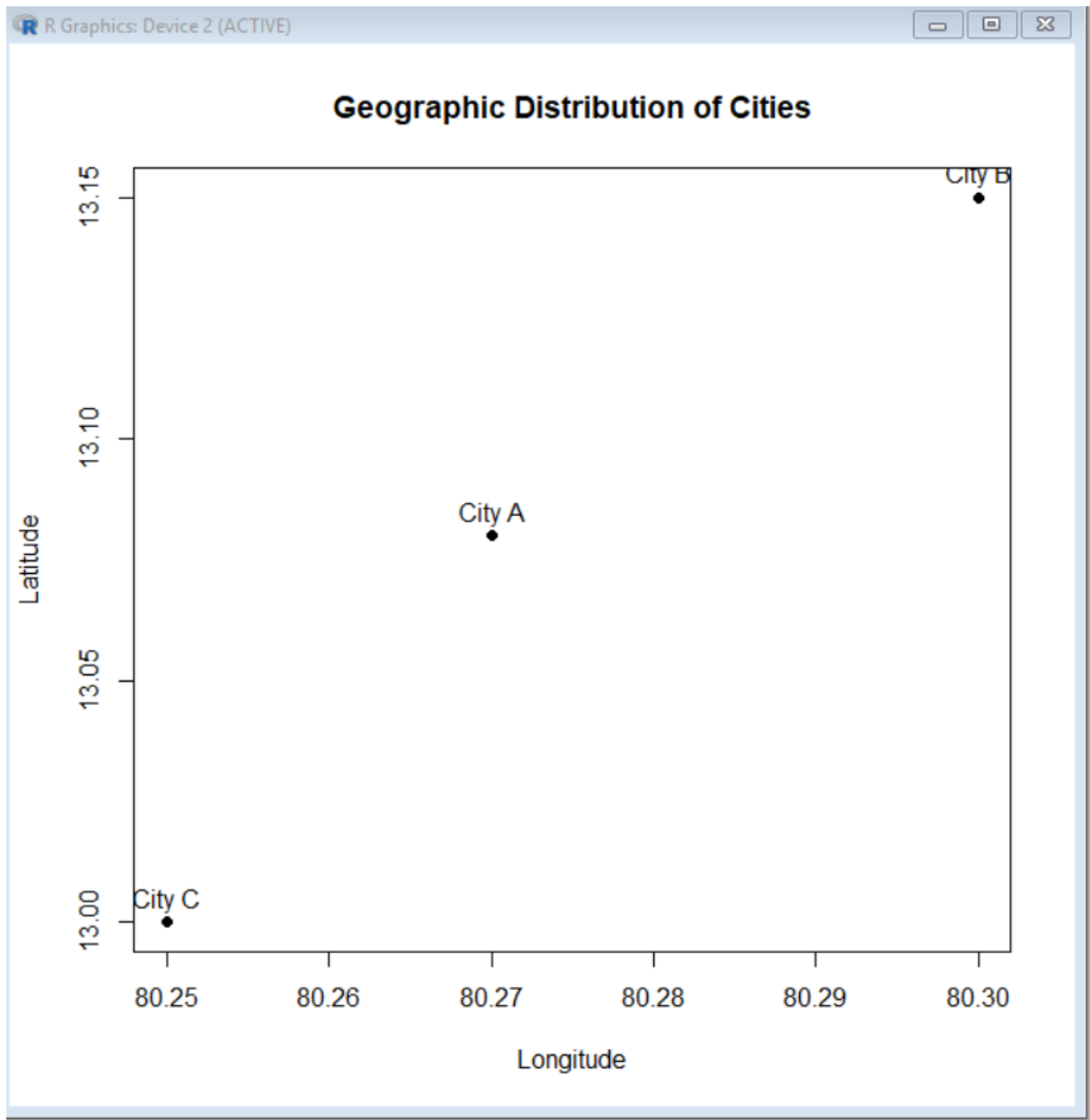
# Create map chart

plot(
  geo_data$Longitude,
  geo_data$Latitude,
```

```
main = "Geographic Distribution of Cities",  
xlab = "Longitude",  
ylab = "Latitude",  
pch = 19  
)
```

```
# Add city labels
```

```
text(  
  geo_data$Longitude,  
  geo_data$Latitude,  
  labels = geo_data$City,  
  pos = 3  
)
```



Observation

The map chart displays the geographic positions of City A, City B, and City C. Each city is labelled clearly on the chart.

Program 2 — Scatter Plot

Aim

To generate a scatter plot to explore the relationship between average temperature and population size.

R Code

```
# Geographic Data
```

```
city <- c(  
  "City A",  
  "City B",  
  "City C"  
)
```

```
population <- c(  
  500000,  
  700000,  
  600000  
)
```

```
temperature <- c(  
  75,  
  68,  
  80  
)
```

```
geo_data <- data.frame(  
  City = city,  
  Population = population,
```

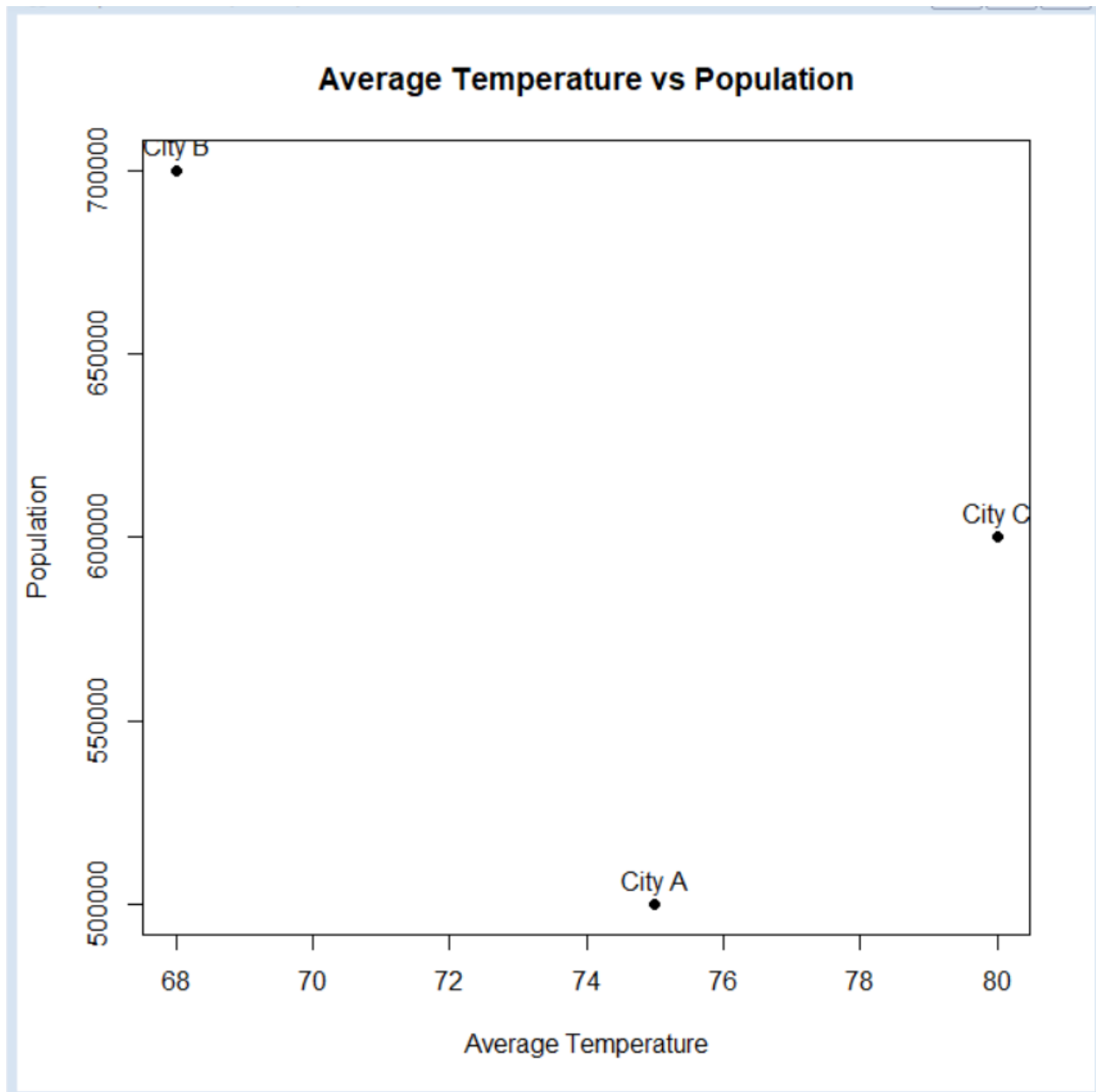
```
Avg_Temperature = temperature
)

# Create scatter plot

plot(
  geo_data$Avg_Temperature,
  geo_data$Population,
  main = "Average Temperature vs Population",
  xlab = "Average Temperature",
  ylab = "Population",
  pch = 19
)

# Add city labels

text(
  geo_data$Avg_Temperature,
  geo_data$Population,
  labels = geo_data$City,
  pos = 3
)
```



Observation / Insights

- City B has the highest population of 700,000.
- City C has the highest average temperature of 80.
- City B has the lowest average temperature of 68.
- The three observations do not show a clear relationship between average temperature and population.

Program 3 — Geographic Data Table

Aim

To build a table displaying the geographic data for each city and label the table elements.

R Code

```
# Geographic Data
```

```
geo_data <- data.frame(  
  City = c(  
    "City A",  
    "City B",  
    "City C"  
  ),  
  Population = c(  
    500000,  
    700000,  
    600000  
  ),  
  Avg_Temperature = c(  
    75,  
    68,  
    80  
  ),  
  Elevation = c(  
    1000,  
    800,  
    1200  
  )  
)
```

```
)  
)
```

```
# Display geographic data table
```

```
geo_data
```

```
>  
> geo_data  
      City Population Avg_Temperature Elevation  
1 City A      5e+05          75         1000  
2 City B      7e+05          68          800  
3 City C      6e+05          80         1200  
>  
>  
,
```

Observation

The table displays the City, Population, Average Temperature, and Elevation for all three cities.

Program 4 — Tableau Interactive Dashboard

Aim

To develop a Tableau dashboard combining the map chart, scatter plot, and table for interactive exploration of geographic data.

Tableau Procedure

Step 1 — Prepare the Dataset

Create an Excel file containing:

City	Population	Avg. Temperature	Elevation
------	------------	------------------	-----------

City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Open Tableau and connect the Excel file.

Step 2 — Create Map Chart

1. Open a new worksheet.
2. Drag City to the Detail field in the Marks card.
3. If Tableau recognizes the city names as geographic locations, select the appropriate Geographic Role.
4. Drag City to the view.
5. Select the Map option.
6. Drag Population to Size.
7. Drag City to Label.
8. Enable Show Mark Labels.
9. Rename the worksheet as: Geographic Map

Step 3 — Create Scatter Plot

1. Create a new worksheet.
2. Drag Avg. Temperature to Columns.
3. Drag Population to Rows.
4. Drag City to Detail.
5. Drag City to Label.
6. Select the circle/scatter visualization.
7. Enable mark labels.
8. Rename the worksheet as: Temperature vs Population

Step 4 — Create Geographic Data Table

1. Create a new worksheet.
2. Drag City to Rows.
3. Drag Population to Text.
4. Drag Avg. Temperature to Text.
5. Drag Elevation to Text.
6. Format the values if required.
7. Rename the worksheet as: Geographic Data Table

Step 5 — Create Tableau Dashboard

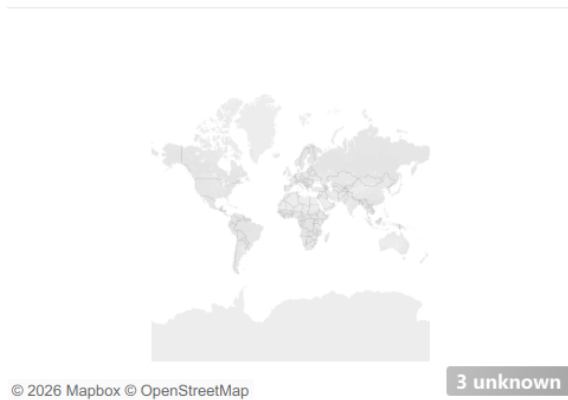
1. Click New Dashboard.
 2. Set the dashboard size according to your requirement.
 3. Drag Geographic Map onto the dashboard.
 4. Drag Temperature vs Population onto the dashboard.
 5. Drag Geographic Data Table onto the dashboard.
 6. Arrange the three visualizations neatly.
 7. Add the dashboard title: Geographic Data Analysis Dashboard
-
8. Enable labels for the map and scatter plot.
 9. Add filters if required.
 10. Save the Tableau workbook.

Dashboard Components

Geographic Data Analysis Dashboard

- **Map Chart** → Geographic distribution of cities
- **Scatter Plot** → Average temperature vs population
- **Table** → City-wise geographic data

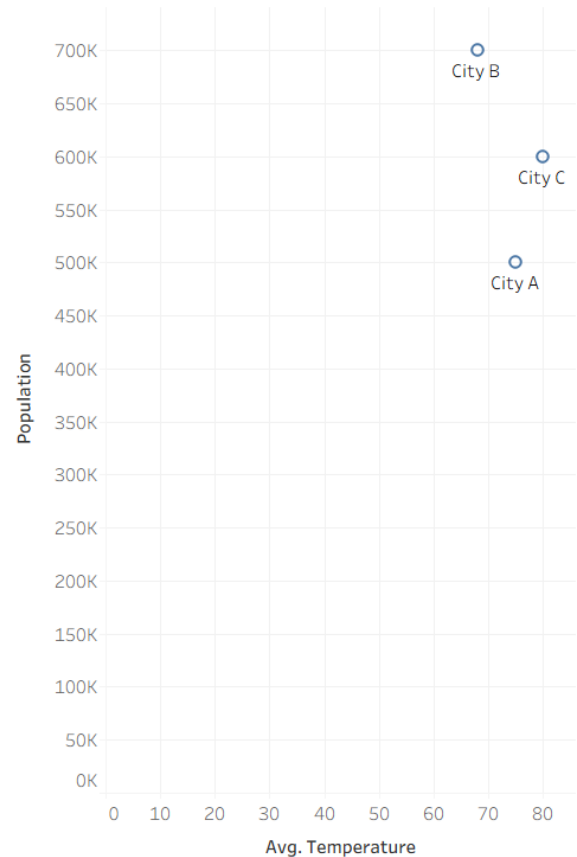
Map Chart



Geographic
Data Table

City	
	5,00,000
City A	75
	1,000
	7,00,000
City B	68
	800
	6,00,000
City C	80
	1,200

Scatter Plot



Observation

The Tableau dashboard combines the map chart, scatter plot, and geographic data table into a single interactive interface. It allows users to explore city locations, population, temperature, and elevation together.

C. Overall Procedure

1. Create the geographic dataset.
2. Enter the City, Population, Average Temperature, and Elevation values.
3. Create a map chart using geographic information.
4. Label the cities on the map.
5. Create a scatter plot between average temperature and population.

6. Analyze the relationship between temperature and population.
7. Create a table containing the geographic information.
8. Import the dataset into Tableau.
9. Create the geographic map in Tableau.
10. Create the temperature-population scatter plot.
11. Create the geographic data table.
12. Combine all three worksheets into a Tableau dashboard.
13. Add labels and interactive features.
14. Save and present the completed dashboard.

D. Result

The geographic data was successfully analyzed using R and Tableau. A map chart, scatter plot, and geographic data table were created, and these visualizations were successfully combined into an interactive Tableau dashboard.

The analysis shows that City B has the highest population of 700,000, while City C has the highest average temperature of 80 and elevation of 1200.

E. Key R Functions Used

Function	Purpose
c()	Creates vectors
data.frame()	Creates the dataset
plot()	Creates map-style and scatter plots
text()	Adds labels to charts
as.Date()	Converts data into date format when required

Key Tableau Features Used

Tableau Feature	Purpose
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Connect to Excel	Imports the dataset
Map	Displays geographic distribution
Columns	Places temperature on the x-axis
Rows	Places population on the y-axis
Marks – Detail	Separates cities
Marks – Label	Displays city names
Marks – Size	Represents population
Dashboard	Combines all visualizations
Filters	Enables interactive exploration